Transformer-based power system energy prediction model

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***Abstract*—With the rapid development of smart grid construction, building energy consumption prediction is gaining more and more attention in energy planning, management, and conservation. Improving the accuracy of energy consumption judgment is a key factor to ensure efficient operation of the energy system. In addition, the model needs to be able to quickly adapt to changes in energy consumption and respond to various emergencies. This paper thus proposes a modified Transformer model based on Multi-head attention and position encoding mechanism. This model makes Transformer focus on the important information which influence the prediction performance by using self-attention mechanism. Compared with LSTM, it achieves the higher prediction accuracy with less training time by the combination of Transformer and position encoding. The final experiment results demonstrate that the proposed method has a good performance in the electronic load prediction task, and it is also suitable for the application scenarios in periodic change prediction such as forecasting sales and price prediction.**

Keywords— Transformer, Multi-head attention, position encoding, energy consumption prediction

I. INTRODUCTION

For many years, the rapid development of modern industry has intensified the consumption of non-renewable energy, and the environmental problems caused by it have gradually increased [1]. In order to ensure sustainable development, efficient and accurate use of energy is required, and the construction of smart grids has been put on the agenda. Among them, the building and construction sector accounted for 36% of global energy consumption, and directly or indirectly produced 40% of carbon dioxide gas [2]. At the same time, the development of Wireless Sensor Networks (WSNs), the Internet of Things (IoT) and other technologies has provided a lot of information about building energy consumption, providing a solid foundation for building energy prediction [3]. Therefore, the prediction of building energy consumption is a powerful way to solve the problem of energy consumption. The development of building energy prediction systems using big data has gradually become one of the research hotspots.

At present, building facility managers face many technical challenges and need to comprehensively consider multiple factors to predict building energy consumption, such as outdoor temperature and humidity, personnel perception behavior, equipment operating conditions, and thermal performance of building envelopes. Because of its extensive and complex influencing factors and significant data fluctuations, the predicted values obtained by using a simple model deviate greatly from the actual values. In recent years, researchers have mainly used machine learning methods to predict building energy consumption, including Support Vector Machines (SVM) [4,5] and artificial neural networks [6-8]. Ma et al. used SVM to predict buildings in South China and used grid search to determine two parameters of support vector machines. At the same time, multiple economic factors such as GDP and household consumption were selected as inputs. Finally, the comparison with the real value verified the practicability of the model [5]; M. A. Rafebis et al. Used the Levenberg-Marqueurd and OWO-Newton hybrid algorithms to optimize the BP neural network model and predict the energy consumption data collected by the TxAIREE research laboratory. The Levinberg-Marques algorithm was used to optimize the final neural network, and the OWO-new independent network was optimized. The optimization results are closer to the real values, but their model goodness of fit is generally lower [6]. Zheng et al. Used a Long Short-Term Memory (LSTM) neural network to build a multi-step load forecasting model and analyzed the spatial impact on the accuracy of household electricity consumption forecasting. The results show that, compared with the prediction directly at the household level, the prediction of equipment and meters electricity consumption, and then the aggregated prediction can help improve the accuracy. However, it brings a sharp increase in the amount of data, and the model requires specific adjustments to different types of equipment [7].

In this paper, we proposed a modified Transformer model based on Multi-head attention and position encoding mechanism. By combining self-attention mechanism and Transformer, the contributions are summarized as follows:

- We successfully applied Transformer model into the power system energy prediction task and greatly improve the prediction accuracy compared with other methods such as LSTM and ARIMA.
- A modified version of Transformer was proposed which change the output layer and the attention layer in the original structure. Proposed model is suitable for regression-related tasks such as time series analysis. Besides, the training time is significantly reduced since the parallel nature in Transformer model.
- We used attention mechanism and position encoding in Transformer which help capture the most

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important features to the prediction performance and remember the periodic patterns in the dataset.

II. BACKGROUND

LSTM solve the problem that traditional neural network cannot remember information from the past and greatly improve the ability of neural network for dealing with long-distance dependency. However, two problems also raise, which are:

- The computation of information in time point t must rely on the results in t-1, which limits the power of parallel computing.
- The information is processed in a sequential way and some information in the early stage may be lost.

Besides, LSTM still cannot cope with the situation with extremely long-distance dependency. The problems above can be solved by following Transformer model, which can be performed parallelly and reduce the distance of two unit into a constant using self-attention. Next we will introduce attention mechanism and the structure of Transformer.

A. Attention Mechanism

Attention mechanism is proposed to substitute LSTM based models and widely used in computer vision. The idea is inspired by the visual processing mechanism of humanity. Briefly, human brain will not process all information after receiving visual input but focus on specific parts. This mechanism has been broadly used in many fields, such as speech recognition, image title generation, machine translation and text classification [9].

In deep neural networks, attention could be seen as a resource allocation method, which distributes more weights or computational resources to information with more essentiality. In summary, this mechanism is useful to solve information overload problem. In practice, attention mechanism can be classified into two different classes: The first is top-down type which is usually conscious and related to specific task. Top-down type often focuses on an object actively and can be set manually. The other is the bottom-up type which is unconscious and has nothing to do with specific tasks. Bottom-up type is driven by the outside and automatically adjusted. So, this type of attention is also known as saliency-based attention. For instance, in LSTM and convolutional neural networks (CNN), gating mechanism which controls the pass of previous information and pooling layer are saliency-based attention.

Here, we use column vectors X to represent input data and lower subscript sequence $[x_1, x_2, \ldots, x_N]$ to represent task-related input vectors. In order to assign proper weighting to specific part of data sequences, attention mechanism leads a query vector q. This query vector calculates the correlation between itself and the input vector by a predefined score function. Besides, an attention variable $t \in [1, N]$ is used to indicate selected position of the input sequence. Detailed processing is shown as follows:

$$\begin{aligned} a_i &= p(t=1 \mid X, q) \\ &= \text{softmax}(s(x_i, q)) \\ &= \frac{\exp(s(x_i, q))}{\sum_{j=1}^{N} \exp(s(x_i, q))} \end{aligned} \tag{1}$$

where α_i is the attention distribution and $s(x_i, q)$ is the score function. In this equation, we can see that a score function needs to be defined. Here, we select Scaled Dot-Product operation to represent score function which is often used in self-attention model. The definition of Scaled Dot-Product is described below:

$$s(x_i, q) = \frac{x_i^T q}{\sqrt{d}} \tag{2}$$

where d represents the dimensionality of input vector. Scaled Dot-Product model comes from the Dot-Product and is normalized over the dimensionality. We can see the difference between two equation lies in the denominator. The former, Scaled Dot-Product is divided by the square root of d. It is easy to see that when d is too large, Dot-Product can have a large variance leading to a small gradient for the SoftMax. To solve this problem, Scaled Dot-Product model is proposed. In summary, self-attention model introduce three vectors; Q, K and V. Q and the Scaled Dot-Product model are utilized as the score function which can generate different weights dynamically and can also be used to deal with sequences containing varied length.

B. Transformer

Transformer model improves the attention mechanism and discards the gated model such as LSTM, which can be seen as a variant of seq2seq based models and consists of encoder and decoder. We will briefly introduce the principles of Transformer below.

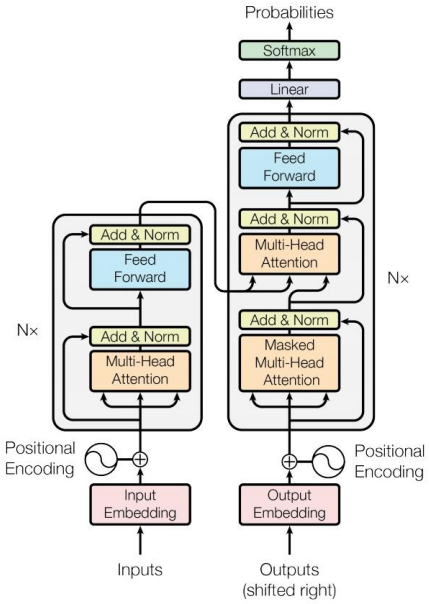


Fig. 1. The structure of original Transformer model, The left part is encoder and right part is decoder.

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Form Fig.1 we can see that there are two new methods used in the Transformer model, which are Multi-Head Attention and Positional Encoding. Multi-Head attention just combines several attention layers allowing them running in parallel and each attention is called a "head". In the experiment, we found that three heads are beneficial for the model performance. The usage of multi-head allows the model to jointly attend to information from different representation subspaces at different positions. In summary, multi-head attention can be formulated as:

$$MultiHead(Q,K,V) = Concat(head_1,\cdots,head_n)W^o$$
$$where\ head_i = Attention(QW_i^Q, KW_i^K, VW_i^V) \quad (3)$$

Where the parameters matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times dq}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^o \in \mathbb{R}^{hd_v \times d_{\text{model}}}$. In this work, we employ h=3, which is the number of heads and $d_k = d_v = \frac{d_{\text{model}}}{h} = 12$. Since the dimension of each head is reduced, the total computational cost is similar to a single head attention layer with full dimension. Masked Multi-head Attention is only used in the decoder since the decoding process is sequential and the decoder only know 0~k-1 elements when it decodes kth element, so by this way it is called "Masked" Multi-head attention. Since the model contains no convolution and recurrence unit, to make the model learn the order of sequence, the information of sequence position must be utilized. Here, we use sine function to encode the position information:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{1000^{\frac{2i}{d_{\text{model}}}}}\right) \quad (4)$$

Where pos is the position and i is the dimension. So we can see that each dimension of the position encoding corresponds to a sinusoid and the position encoding has the same dimension d_{model} as the embeddings, thus the two can be summed.

III. PROPOSE METHOD

A. Design

Our proposed model is illustrated in the Fig.2, besides the historical electronic load, some external factors are also considered, including weather condition, day type and building features. All the external factors are combined with historical load as input to the Transformer and other benchmark methods. Since the original Transformed was designed for text analysis, we modify the structure to make it suitable for regression analysis. For the output layer, we change the SoftMax into single neuron and use summation as output. To accelerate the training process, we remove the Multi-head attention part in the decoder since one attention layer is enough for the prediction task. Before the experiment, we need to preprocess the dataset since LSTM and Transformer model are very sensitive to the data scale. We use MinMax to normalize the dataset and restrict the value between [0,1]. The formula is described as below:

$$x_{norm} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (5)$$

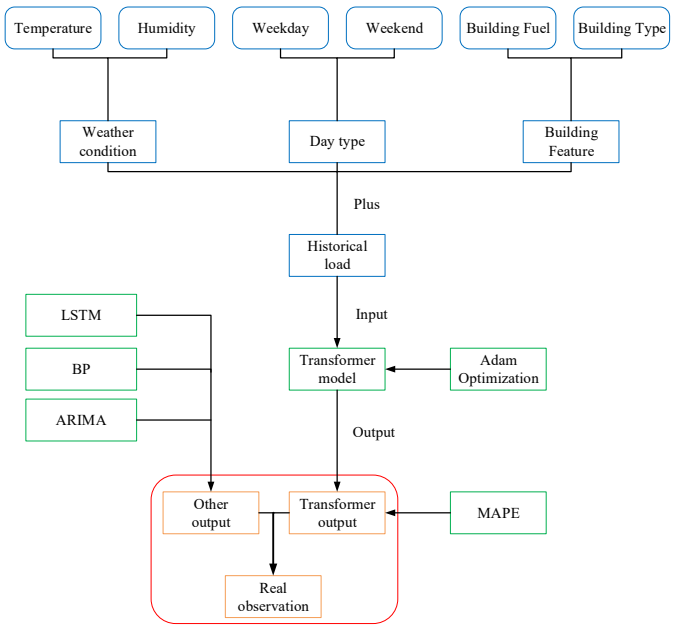


Fig. 2. Structure of proposed Transformer model and benchmarks for building electronic load prediction.

where x_{norm} indicates the value after normalization, $x_{\max}$ and $x_{\min}$ are the maximum and minimum value in the dataset respectively. To evaluate the performance, we take the measure Mean Absolute Percentage Error (MAPE) as the evaluation criterion, which can be computed by the formula:

$$\varepsilon_{mape} = \frac{1}{n}\sum_{i=1}^{n}\frac{|y_i - y_l|}{y_i} \quad (6)$$

where y_i and y_l are the real value and prediction value respectively. The dataset is divided into two parts, training set and testing set, in which the number of training set accounts for 80% and testing set accounts for 20%.

B. Adam

In this paper, Adam optimization algorithm is used to train our model [10]. Adam belongs to first-order optimization group which is based on stochastic gradient descent (SGD). A flaw of SGD is that the learning rate does not change during the training process which lowers the training efficiency. So Adam use a changing learning rate during the training process. It estimates the rate by calculating the first order and second-order moments of the gradient. This idea is groundbreaking, and it is the first adaptive learning algorithm. What's more, Adam combines the advantages of AdaGrad and RMSProp. The training procedure of Adam is summarized in Fig.3. In the experiment part we will show that Adam algorithm has excellent performance and the model achieves a MAPE lower than 1.0 in the prediction. After using Adam algorithm, the loss on training set and test set can be reduced to about 0.042.

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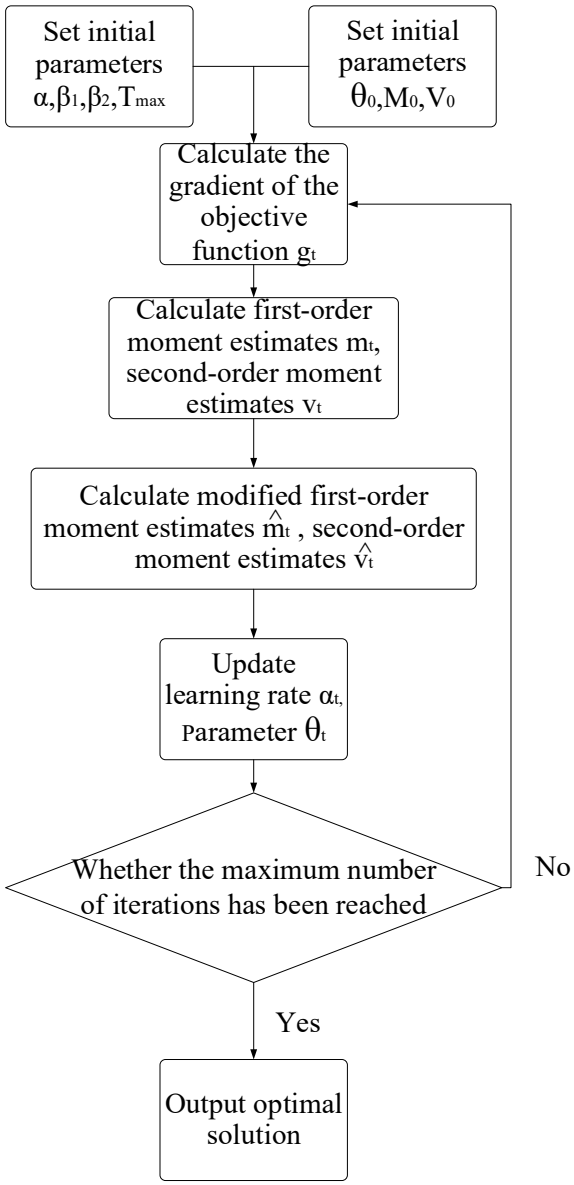


Fig. 3. Adam algorithm process.

IV. EXPERIMENT AND RESULT

A. Dataset

In the experiment, we use a dataset from UC Irvine School of Information and Computer Science to test our model. This dataset contains hourly load profile data for 16 commercial building types and residential buildings. Besides, the dataset uses the Residential Energy Consumption Survey (RECS) for statistical references of building types by location. This dataset is approximately 4.8GiB compressed or 19GiB uncompressed and Residential High and Low load files contains data from 366 days in a year for leap years to the more general 365 days in a normal year. To make accurate prediction, some periodic factor should also be considered, such as temperature, humidity, building type and season.

TABLE I. SUMMARY OF BUILDING FUEL TYPES AND STRUCTURE TYPES IN DATASET.

	Selected Building Fuel Types				
	Very Cold/Cold	***Mixed-Humid***	***Mixed-Dry/Hot-Dry***	***Hot-Humid***	***Marine***
Space Heating	Natural Gas	Natural Gas	Natural Gas	Electric	Natural Gas
Air Conditioning	Yes	Yes	Yes	Yes	No
Water Heating	Natural Gas	Electric	Natural Gas	Electric	Natural Gas
	Selected Building Structure Types				
	Very Cold/Cold	***Mixed-Humid***	***Mixed-Dry/Hot-Dry***	***Hot-Humid***	***Marine***
Total Size (sq. ft.)	2696	2546	2000	2023	2090
Urban and Rural	Urban	Urban	Urban	Urban	Urban
Metropolitan and Micropolitan	Metro	Metro	Metro	Metro	Metro
Number of Stories / Levels	1 Story	1 Story	1 Story	1 Story	1 Story
Major Outside Wall Construction	Siding (Aluminum, Vinyl, Steel)	Siding (Aluminum, Vinyl, Steel)	Stucco	Brick	Wood
Major Roofing Material	Ceramic or Clay Tiles	Ceramic or Clay Tiles	Ceramic or Clay Tiles	Ceramic or Clay Tiles	Ceramic or Clay Tiles
Foundation/Basement of Single-Family	Basement	Concrete Slab	Concrete Slab	Concrete Slab	Crawl space
Bedrooms	3	3	3	3	3
Full Bathrooms	1	1	2	2	1
Half Bathrooms	None	None	None	None	None
Basement Single-Family Homes	Yes	No	No	No	No
Finished Basement	No	No Basement	No Basement	No Basement	No Basement
Type of Glass in Windows	Double-pane Glass	Double-pane Glass	Single-pane Glass	Single-pane Glass	Double-pane Glass

Some important characteristics of building in the dataset are listed in Tab.1.

The configurations of experimental platform are listed as below:

- Operation system: windows 10
- CPU: Intel Core i9-9900X
- RAM: 32 GB
- Deep learning framework: TensorFlow 1.13.1
- Developing tool: Visual Studio Code
- Programming language: Python 3.6

B. Experiment Result

After 4 epochs the algorithm almost converge and achieve a MAPE under 1.0 on the test set. To evaluate the

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model performance, we benchmark our method by classic Autoregressive Integrated Moving Average model (ARIMA), BP neural network and LSTM. The results are shown below:

TABLE Ⅱ. MEAN ABSOLUTE PERCENTAGE ERROR OF EACH MODEL.

MAPE(%)	weekday	weekend	Whole data
Transformer	0.97	1.12	1.08
LSTM	1.24	1.67	1.43
BP	7.55	23.54	17.90
ARIMA	5.77	12.61	8.43

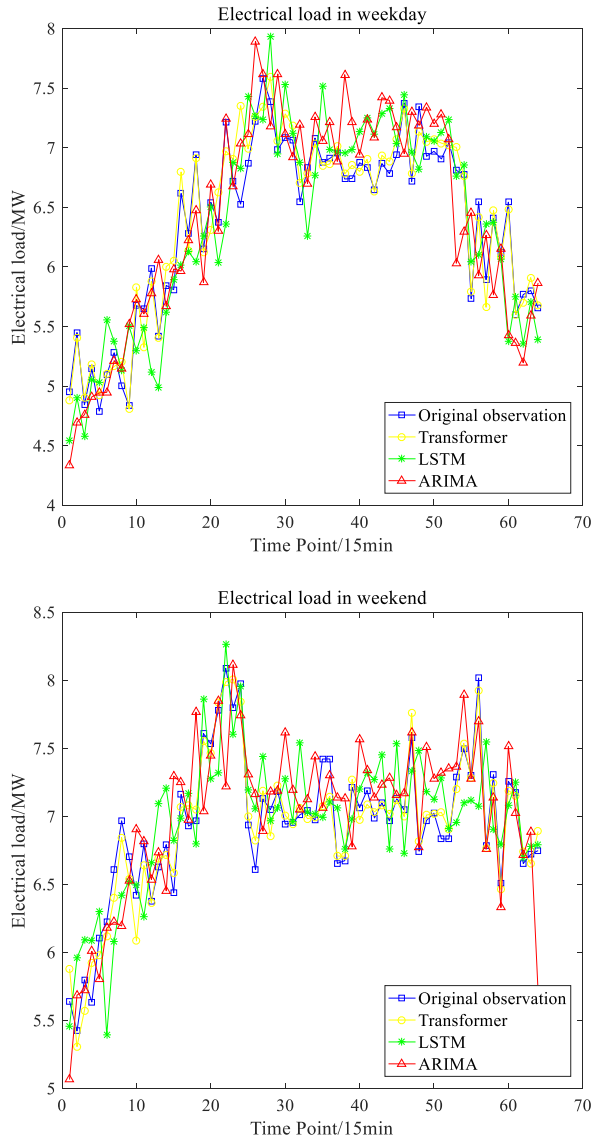


Fig. 4. The building electronic load prediction in weekday and weekend. The start time is 06:00 AM and the end time is 10:00 PM. We make a prediction every 15 minutes and the solid line is the real load value.

From Tab.2 we can see that Transformer model achieve smallest MAPE among test set including weekday and weekend. We also illustrate the prediction line of the four models in Fig.4 and Transformer model fits well into the real load value. In a word, compared with ARIMA and LSTM, the prediction accuracy is greatly improved when using Transformer model. Also, the time consuming of training Transformer is greatly reduced by the parallel nature of Multi-head attention mechanism. In this experiment, GPU is not used to accelerate the calculation, and the CPU time consuming is about 35min.

V. CONCLUSION

This paper presents a modified Transformer model based on Multi-head attention and position encoding mechanism. Compared with LSTM, it achieves the higher prediction accuracy with less training time. By using self-attention mechanism, Transformer can focus on the important information which influence the prediction performance mostly. Furthermore, the position encoding can help the model remember the latent pattern in different time point, especially for the periodic data. The combination of these two methods exhibits outstanding capability in regression task such as time series analysis. The experimental results show that the model has a good performance in the electronic load prediction task, and it is also suitable for the application scenarios in periodic change prediction such as forecasting sales and stock price prediction.

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